THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 24	Find the hypotenuse
2	**(c)** 8	Midpoint
3	**(c)** 4	$\frac{x_1+x_2+x_3}{3}$
4	**(c)** 5	Distance from centre
5	**(d)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$
6	**(d)** 10	Area $= 0$
7	**(b)** 30	Find the hypotenuse
8	**(a)** 6	Midpoint
9	**(d)** 3	$\frac{x_1+x_2+x_3}{3}$
10	**(b)** 13	Distance from centre

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(6)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 64$ (1); $y = 2 \pm 8$, i.e. $y = 10$ or -6 (1).	1+1+1
12	Diagonals bisect each other: midpoint of AC is $\left(\frac{11}{2}, \frac{7}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (5, 5)$ (2).	2+1+2
13	Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (6, 4)$ (2).	1+2
14	$(6)^2 + (y+1)^2 = 100$ (1); $(y+1)^2 = 64$ (1); $y = -1 \pm 8$, i.e. $y = 7$ or -9 (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 16$, $BC^2 = 13$, $CA^2 = 13$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
16	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{2(8)+3(-2)}{5} = 2$ (1); $y = 7$ (1).	1+1+1
17	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -12 + 12 + 32\rvert$ (1); $= 16$ sq. units (1).	1+1+1
	OR: $AB^2 = 16$, $BC^2 = 20$, $CA^2 = 20$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
18	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{3(7)+1(1)}{4} = \frac{11}{2}$ (1); $y = 5$ (1).	1+1+1
	OR: Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -32 - 4 - 36\rvert$ (1); $= 36$ sq. units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let ratio $m : n$; $y = \frac{m(4)+n(-5)}{m+n} = 0$ (2); $m : n = \frac{5}{4} : 1$, i.e. 5:4 (1); point $x = \frac{5(6)+4(-3)}{9} = 2$ (2).	2+1+2
	OR: Let the point be $(x, 0)$; $(x-1)^2 + 25 = (x-7)^2 + 9$ (2); $x = \frac{8}{3}$ (2); point $\left(\frac{8}{3}, 0\right)$ (1).	2+2+1
20	Let ratio $m : n$; $y = \frac{m(7)+n(-8)}{m+n} = 0$ (2); $m : n = \frac{8}{7} : 1$, i.e. 8:7 (1); point $x = \frac{8(4)+7(1)}{15} = \frac{13}{5}$ (2).	2+1+2
	OR: Let the point be $(x, 0)$; $(x+3)^2 + 4 = (x-5)^2 + 36$ (2); $x = 3$ (2); point $(3, 0)$ (1).	2+2+1

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$
2	**(b)** 9	Area $= 0$
3	**(d)** 36	Find the hypotenuse
4	**(c)** 9	Midpoint
5	**(a)** 5	$\frac{x_1+x_2+x_3}{3}$
6	**(c)** 10	Distance from centre
7	**(d)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$
8	**(c)** 11	Area $= 0$
9	**(d)** 40	Find the hypotenuse
10	**(d)** 8	Midpoint

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diagonals bisect each other: midpoint of AC is $(3, \frac{5}{2})$ (2); this is also the midpoint of BD (1); $D = (2,\ 4)$ (2).	2+1+2
12	Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (3,\ 4)$ (2).	1+2
13	$(12)^2 + (y-4)^2 = 169$ (1); $(y-4)^2 = 25$ (1); $y = 4 \pm 5$, i.e. $y = 9$ or -1 (1).	1+1+1
14	Diagonals bisect each other: midpoint of AC is $(\frac{9}{2}, \frac{9}{2})$ (2); this is also the midpoint of BD (1); $D = (4,\ 6)$ (2).	2+1+2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 16$, $BC^2 = 20$, $CA^2 = 20$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
16	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{2(10)+3(0)}{5} = 4$ (1); $y = 6$ (1).	1+1+1
17	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert 0 + 25 + 28\rvert$ (1); $= \frac{53}{2}$ sq. units (1).	1+1+1
	OR: $AB^2 = 36$, $BC^2 = 34$, $CA^2 = 34$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
18	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{3(-5)+2(5)}{5} = -1$ (1); $y = 8$ (1).	1+1+1
	OR: Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -24 + 36 + 20\rvert$ (1); $= 16$ sq. units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let ratio $m : n$; $y = \frac{m(9)+n(-3)}{m+n} = 0$ (2); $m : n = \frac{1}{3} : 1$, i.e. 1:3 (1); point $x = \frac{1(10)+3(-2)}{4} = 1$ (2).	2+1+2
	OR: Let the point be $(x, 0)$; $(x-2)^2 + 64 = (x-10)^2 + 4$ (2); $x = \frac{9}{4}$ (2); point $(\frac{9}{4}, 0)$ (1).	2+2+1
20	Let ratio $m : n$; $y = \frac{m(10)+n(-6)}{m+n} = 0$ (2); $m : n = \frac{3}{5} : 1$, i.e. 3:5 (1); point $x = \frac{3(8)+5(0)}{8} = 3$ (2).	2+1+2
	OR: Let the point be $(x, 0)$; $(x-4)^2 + 25 = (x+2)^2 + 1$ (2); $x = 3$ (2); point $(3, 0)$ (1).	2+2+1

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	$\frac{x_1+x_2+x_3}{3}$
2	**(d)** 15	Distance from centre
3	**(b)** 1	$x = \frac{kc+a}{k+1} = m$
4	**(c)** 12	Area $= 0$
5	**(a)** 56	Find the hypotenuse
6	**(b)** 10	Midpoint
7	**(b)** 5	$\frac{x_1+x_2+x_3}{3}$
8	**(a)** 5	Distance from centre
9	**(a)** $\frac{1}{3}$	$x = \frac{kc+a}{k+1} = m$
10	**(c)** 9	Area $= 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (5,\ 6)$ (2).	1+2
12	$(5)^2 + (y+0)^2 = 169$ (1); $(y+0)^2 = 144$ (1); $y = 0 \pm 12$, i.e. $y = 12$ or -12 (1).	1+1+1
13	Diagonals bisect each other: midpoint of AC is $\left(\frac{9}{2}, \frac{13}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (4,\ 8)$ (2).	2+1+2
14	Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (2,\ 3)$ (2).	1+2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 36$, $BC^2 = 45$, $CA^2 = 45$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
16	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{1(-8)+3(4)}{4} = 1$ (1); $y = 4$ (1).	1+1+1
17	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -12 - 30 - 18\rvert$ (1); $= 30$ sq. units (1).	1+1+1
	OR: $AB^2 = 36$, $BC^2 = 34$, $CA^2 = 34$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
18	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{5(6)+1(-6)}{6} = 4$ (1); $y = 9$ (1).	1+1+1
	OR: Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -5 + 49 - 8\rvert$ (1); $= 18$ sq. units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let ratio $m:n$; $y = \frac{m(8)+n(-4)}{m+n} = 0$ (2); $m:n = \frac{1}{2}:1$, i.e. 1:2 (1); point $x = \frac{1(5)+2(2)}{3} = 3$ (2).	2+1+2
	OR: Let the point be $(x, 0)$; $(x+0)^2 + 16 = (x-6)^2 + 4$ (2); $x = 2$ (2); point $(2, 0)$ (1).	2+2+1
20	Let ratio $m:n$; $y = \frac{m(5)+n(-7)}{m+n} = 0$ (2); $m:n = \frac{7}{5}:1$, i.e. 7:5 (1); point $x = \frac{7(3)+5(-1)}{12} = \frac{4}{3}$ (2).	2+1+2
	OR: Let the point be $(x, 0)$; $(x-3)^2 + 36 = (x-11)^2 + 16$ (2); $x = \frac{23}{4}$ (2); point $(\frac{23}{4}, 0)$ (1).	2+2+1

Mistakes to avoid:

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